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Classification Lecture 8

Saturday, 5 October 2024 8:11 AM

Logistic Regression

↳ predictive technique used for classification problems.

Unlike linear regression which outputs continuous numbers, logistic regression tranforms its output using the logistic sigmoid function to return a probability value which can be mapped to two or more discrete classes.

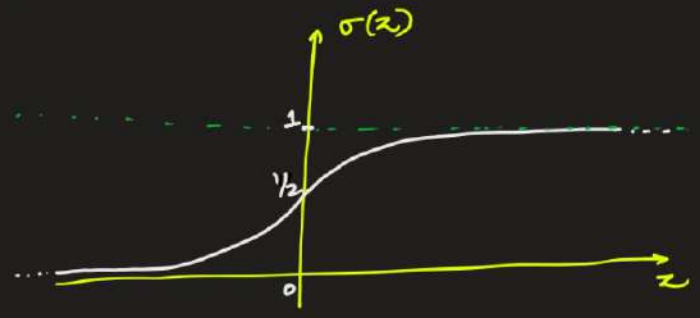
Linear Regression :- Predicts a continuous outcome based on the sum of the weighted inputs. It fits a line to data points

Logistic Regression :- Predicts a categorical outcome. Insted af fitting a line, it fits an "S-shaped" logistic function. known as the sigmoid function. The output of the sigmoid function is used to categorize the data into classes (usually 0/1)

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

↑

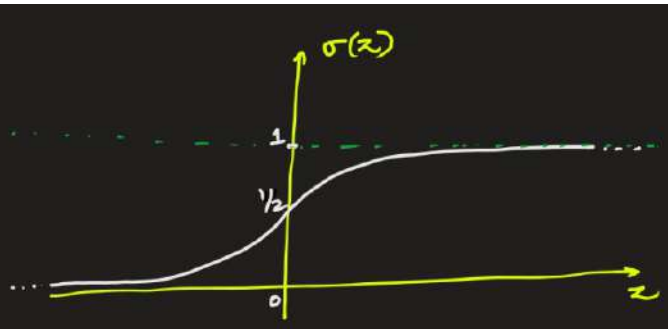
Sigmoid function



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$$\sigma(z) = \frac{1}{1+e^{-z}}$$

↑ Sigmoid function



- The output is bounded between 0 and 1, making it suitable for modeling a probability

Mathematical Formulation of Logistic Regression

Odds: In probability, is the ratio of the probability that the event will occur to the probability that it will not occur

$$\text{Odds} = \frac{P(\text{event})}{1 - P(\text{event})}$$

Log odds: The log odds is simply the natural logarithm (log base e) of the odds

$$\text{Log odds} = \log_e\left(\frac{P(\text{event})}{1 - P(\text{event})}\right)$$

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Model Formulation: Logistic regression models the log odds of the dependent binary outcome as a linear relationship with the independent variables (predictors)

$$\log\left(\frac{P(Y=1|X)}{1-P(Y=1|X)}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$$

Each unit increase in x_k multiplies the odds of the outcome by e^{β_k}

Suppose that the model is predicting the likelihood of a disease based on age, the coefficient of age is $\beta_{age} = 0.05$.

(For every additional year of age, the odds of having a disease are multiplied by $e^{0.05} \approx 1.05$, indicating a 5% increase in the odds each year.

$$\Rightarrow \log\left(\frac{P(Y=1|X)}{1-P(Y=1|X)}\right) = \beta_0 + \beta_1 x_1 + \dots + \beta_{age} x_{age} + \dots$$

$$\text{odds} \propto e^{\beta_{age} \cdot x_{age}} = e^{0.05 \times x_{age}}$$

$$e^{0.05} \rightarrow e^{0.05 \times 2} \rightarrow e^{0.05 \times 3}$$

multiplying $e^{0.05} \approx 1.05$

$$\log\left(\frac{P(Y=1|X)}{1-P(Y=1|X)}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$$

$$= \underset{1\times(p+1)}{\vec{X}} \; \underset{(p+1)\times 1}{\vec{\beta}}$$

$$= \underset{1\times 1}{z}$$

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$$\therefore \log\left(\frac{p}{1-p}\right) = z \quad \Rightarrow \quad \frac{p}{1-p} = e^z$$

$$\Rightarrow p = e^z - e^z p$$

$$\Rightarrow (1+e^z)p = e^z$$

$$\Rightarrow p = \frac{e^z}{1+e^z} = \frac{e^{x\beta}}{1+e^{x\beta}}$$

$$= \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-x\beta}}$$

$$= \sigma(z) = \sigma(x\beta)$$

$$= \sigma(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p)$$

$$\boxed{P(Y=1 \mid X = x_1, x_2 \cdots x_p) = \sigma(\beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p)}$$

$$\therefore P(Y=0 \mid X = x_1 x_2 \cdots x_p) = 1 - \sigma(z)$$

Maximum Likelihood Estimation for Logistic Regression

$\beta_0, \beta_1 \cdots \beta_p$

$P(Y=1 \mid x) = \sigma(z)$

$P(Y=0 \mid x) = 1 - \sigma(z)$

} Bernoulli distribution

Objective:- Find the optimal values of $\vec{\beta}$ given a labelled training dataset on n data points $(x_1, y_1), (x_2, y_2) \cdots (x_N, y_N)$

					Y (0/1)
→ X_1	$(x_{11}$	x_{21}	$\cdots$	$x_{p1})$	0
X_2	x_{12}	x_{22}	$\cdots$	x_{p2}	1
⋮	⋮	⋮		⋮	⋮
X_N	x_{1N}	x_{2N}	$\cdots$	x_{pN}	0

$P(Y=1 \mid x) = \sigma(z)$

what β maximizes the likelihood of the sample?

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For $y=1$, $P(x;\beta) = P(x)$

For $y=0$, $P(x;\beta) = 1-P(x)$

$$L(x_1;\beta) = P(x_1)^{y_1} \cdot (1-P(x_1))^{1-y_1}$$

↓ we have n such independent points

$$L(x;\beta) = \prod_{i=1}^{N} P(x_i)^{y_i} \cdot (1-P(x_i))^{1-y_i}$$

$$\log L(x;\beta) = \sum_{i=1}^{N} \left[y_i \log P(x_i) + (1-y_i) \log (1-P(x_i)) \right]$$

$$= \sum_{i=1}^{N} \log (1-P(x_i)) + \sum_{i=1}^{N} y_i \boxed{\log \frac{P(x_i)}{1-P(x_i)}} \rightarrow \text{log odds} = z_i = x_i\beta$$

$$= \boxed{\sum_{i=1}^{N} \left[-\log (1+e^{x_i\beta}) \right] + \sum_{i=1}^{N} y_i \, x_i\beta}$$

$$= -\sum_{i=1}^{N} \log \left(1+e^{\beta_0 + x_{1i}\beta_1 + x_{2i}\beta_2 + \cdots + x_{pi}\beta_p}\right) + \sum_{i=1}^{N} y_i \left(\beta_0 + x_{1i}\beta_1 + \cdots + x_{pi}\beta_p\right)$$

$$\frac{\partial \log L(x;\beta)}{\partial \beta_0} = 0 \quad \cdots \quad \frac{\partial \log L(x;\beta)}{\partial \beta_1} = 0 \quad \cdots \quad \frac{\partial \log L(x;\beta)}{\partial \beta_p} = 0$$

↑ ↑ ↑

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The derivatives of the log-likelihood function do no NOT simplify to a form that can be easily solved for $\vec{\beta}$. The presence of $\sigma(x\beta)$, which includes both $e^{x\beta}$ and its reciprocal, results in transcedental equations that typically don't have closed-form solutions.

We use Numerical optimization techniques like Gradient Ascent or Newton-Raphson method to find the optimal values of β.

$$\log L(x;\beta) = -\sum_{i=1}^{N} \log\left(1+e^{\beta_0+x_{1i}\beta_1+x_{2i}\beta_2+\cdots+x_{pi}\beta_p}\right) + \sum_{i=1}^{N} y_i\left(\beta_0+x_{1i}\beta_1+\cdots+x_{pi}\beta_p\right)$$

$$\frac{\partial \log L(x;\beta)}{\partial \beta_0} = -\sum_{i=1}^{N} \frac{e^{x_i\beta}}{1+e^{x_i\beta}} + \sum_{i=1}^{N} y_i = -\sum_{i=1}^{N}\sigma(z_i) + \sum_{i=1}^{N} y_i$$

$$\frac{\partial \log L(x;\beta)}{\partial \beta_k} = -\sum_{i=1}^{N} \frac{e^{x_i\beta}}{1+e^{x_i\beta}}\cdot x_{ki} + \sum_{i=1}^{N} y_i \cdot x_{ki} = -\sum_{i=1}^{N}\sigma(z_i)\cdot x_{ki} + \sum_{i=1}^{N} y_i x_{ki}$$

$k \neq 0$

$$\frac{\partial \log \text{likelihood}}{\partial \beta_0} = \sum_{i=1}^{N}(y_i - \sigma(z_i)),$$

$$\frac{\partial \log \text{likelihood}}{\partial \beta_k} = \sum_{i=1}^{N}(y_i - \sigma(z_i))\cdot x_{ki}$$

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Gradient ascent to find $\vec{\beta}$ values

① Initialize the parameters $\vec{\beta}$ $(\beta_0, \beta_1 \cdots \beta_p)$
- ↳ zero initialization
- ↳ Random

② Calculate $\frac{\partial \log likelihood}{\partial \beta_k}$; $0 \leq k \leq p$

③ update β values in the direction of gradient

$$\beta_k = \beta_k + \alpha \cdot \frac{\partial L}{\partial \beta_k} \longrightarrow \text{using eqn ①}$$

④ Stop at convergence

For prediction of class of a new point;

$$\log\left(\frac{P(y=1 \mid x=\text{new point})}{1-P(y=1 \mid x=\text{new point})}\right) = z = \underset{\text{new point}}{\vec{x}}\vec{\beta}$$

↳ $P(y=1 \mid x=\text{new point}) = \sigma(z) = \sigma(x\beta)$

$$= \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-x\beta}} \quad \text{↳ new point}$$

$$P(Y=0 \mid X=np) = 1 - P(Y=1 \mid X=np)$$

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Eg #

Person	Income (x)	Purchase? (y) (0/1)
→ 1	40	0
→ 2	55	1
→ 3	30	0
→ 4	70	1

$x = 65$, predict they will purchase or not using logistic regression.

Sol. $z = \beta_0 + \beta_1 x$ $\quad \sigma(z) = \frac{1}{1+e^{-z}}$

$$\frac{\partial \text{loglik}}{\partial \beta_0} = \sum_{i=1}^{N} (y_i - \sigma(z_i)), \quad \frac{\partial \text{loglik}}{\partial \beta_1} = \sum_{i=1}^{N} (y_i - \sigma(z_i)) \cdot x_i$$

Initialization $\beta_0 = 0$, $\beta_1 = 0$, LR, $\alpha = 0.01$

1st iteration :- $\frac{\partial L}{\partial \beta_0} = \left(0 - \frac{1}{1+e^{-0}}\right) + \left(1 - \frac{1}{1+e^{0}}\right) + \left(0 - \frac{1}{1+e^{0}}\right) + \left(1 - \frac{1}{1+e^{0}}\right) = -\frac{1}{2} + \frac{1}{2} - \frac{1}{2} + \frac{1}{2} = 0$

$$\frac{\partial L}{\partial \beta_1} = \left(-\frac{1}{2}\right) 40 + \left(\frac{1}{2}\right) 55 + \left(-\frac{1}{2}\right) 30 + \left(\frac{1}{2}\right) 70 = 27.5$$

$$\beta_0 = \beta_0 + \alpha \cdot \frac{\partial L}{\partial \beta_0} = 0 + 0.01 \times 0 = 0$$

$$\beta_1 = \beta_1 + \alpha \cdot \frac{\partial L}{\partial \beta_1} = 0 + 0.01 \times 27.5 = 0.275$$

2nd iteration :- $\frac{\partial L}{\partial \beta_0} = \left(0 - \frac{1}{1+e^{-(0+0.275 \times 40)}}\right) + \cdots$

$\frac{\partial L}{\partial \beta_1} = \cdots$

$\vdots$

$\boxed{\beta_0}, \boxed{\beta_1}$ $\quad P(y=1 \mid x=65) = \sigma(z) = \frac{1}{1+e^{-(\beta_0 + \beta_1 \times 65)}} \cdots$